

Inversion Probability

Time limit: 1.00 s

Memory limit: 512 MB

An array has n integers $x_1, x_2, \dots, x_n$, and each of them has been randomly chosen between 1 and r_i . An inversion is a pair (a, b) where $a < b$ and $x_a > x_b$.

What is the expected number of inversions in the array?

Input

The first input line contains an integer n : the size of the array.

The second line contains n integers $r_1, r_2, \dots, r_n$: the range of possible values for each array position.

Output

Print the expected number of inversions rounded to six decimal places (rounding half to even).

Constraints

- $1 \leq n \leq 100$
- $1 \leq r_i \leq 100$

Example

Input	Output
3 5 2 7	1.057143